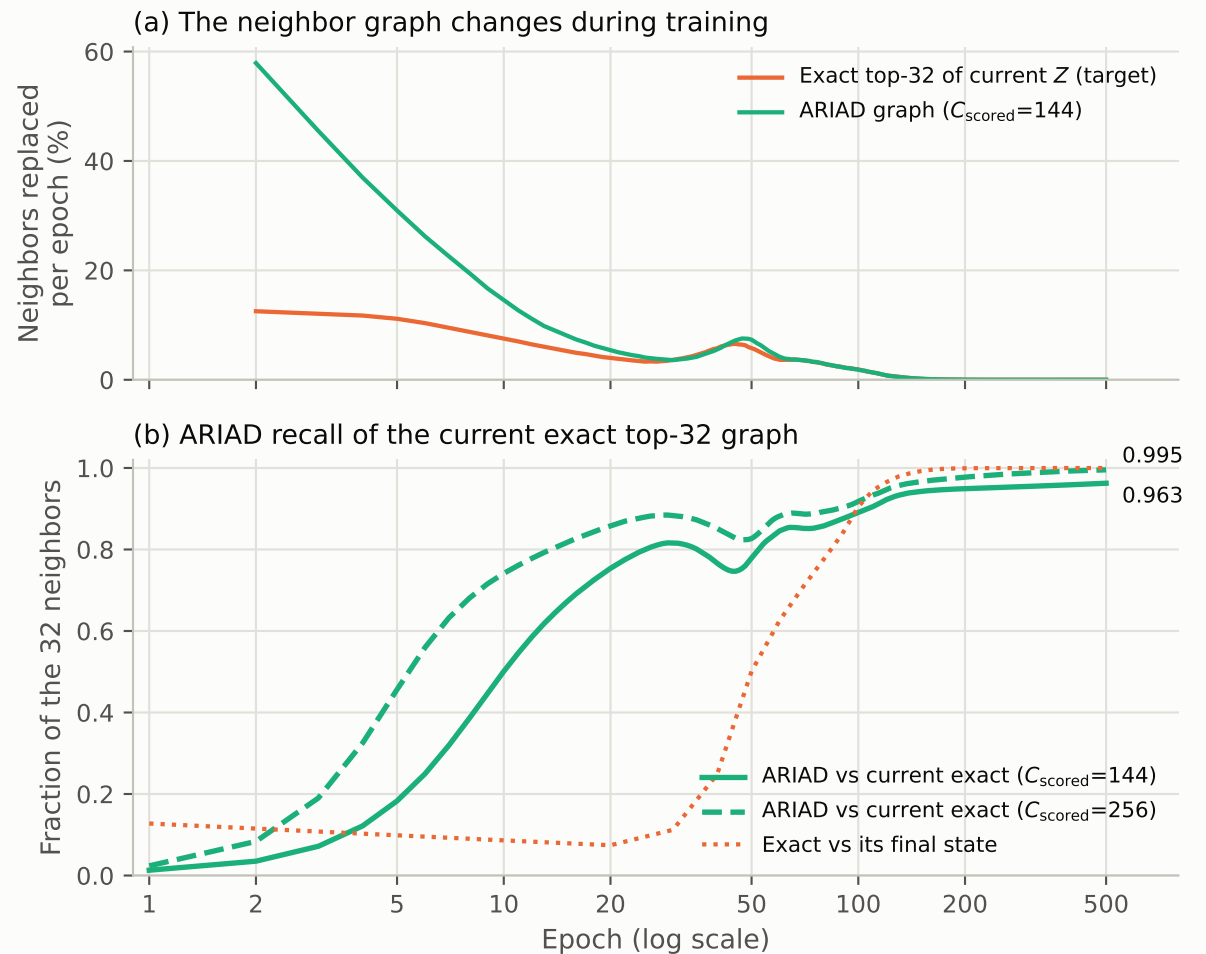


Graph maintenance during training (N=10,000, k=32, sparse top-32 teacher)



Exact audit of all N rows every epoch against the post-step student Z; not part of training cost. Single seed.
Replacement = $1 - |S_t \cap S_{t-1}| / 32$, averaged over rows. ARIAD's early peak is its first refreshes away from the random initial graph. Panel (a) and the dotted line use the $C_{\text{scored}}=144$ run.